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(54) **VARIABLE DIVERGENCE LASER FOR DYNAMIC FOCUS ADJUSTMENT**

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Publication Classification

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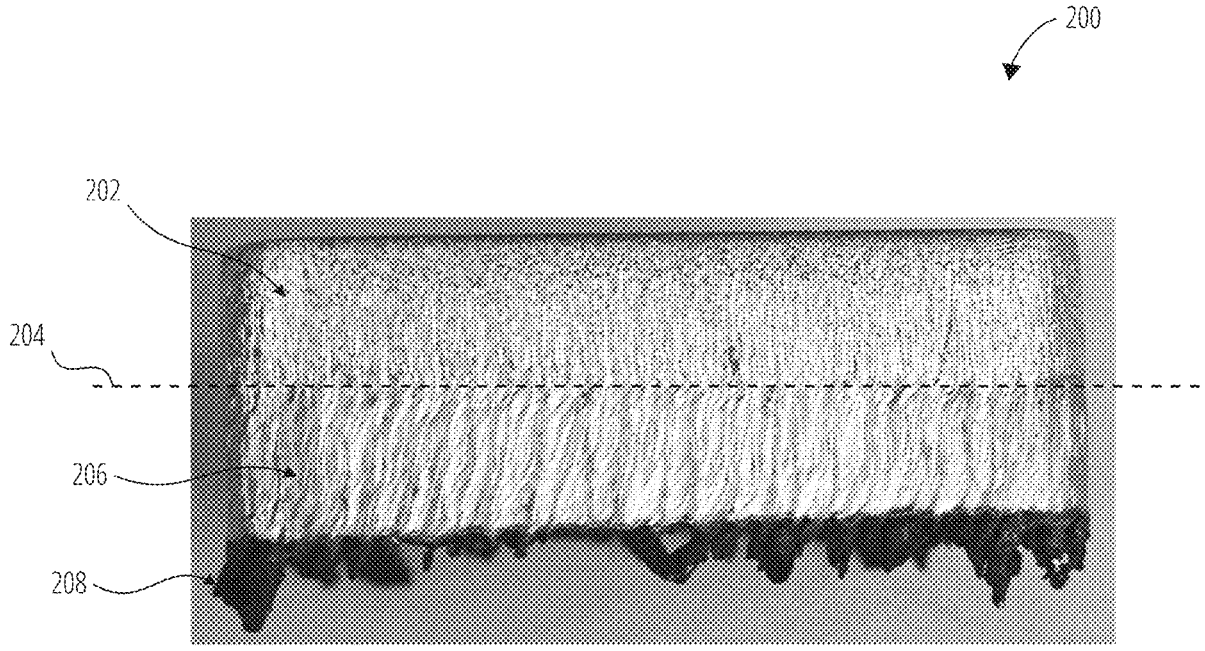
G02B 27/09	(2006.01)
B23K 26/06	(2006.01)
G02B 19/00	(2006.01)

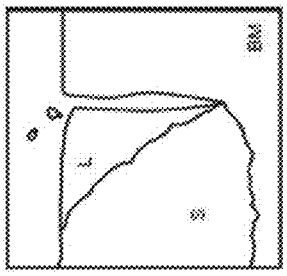
(52) **U.S. Cl.**

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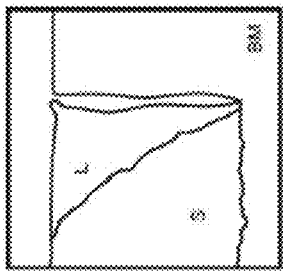
(57) **ABSTRACT**

A laser processing system has a laser source and a divergence-tuning beam characteristic conditioner configured to, in response to a control input, repetitively change a variable divergence laser beam between a first divergence and a second divergence that is different from the first divergence. The system also includes a process head having a collimating lens and a focusing lens. And the system includes a delivery fiber coupled to the divergence-tuning beam characteristic conditioner for guiding the variable divergence laser beam and launching it to the process head. The collimating lens is configured to receive the variable divergence laser beam and direct it as an intermediate beam to the focusing lens to focus the intermediate beam toward the workpiece at a first depth corresponding to the first divergence and at a second depth corresponding to the second divergence.

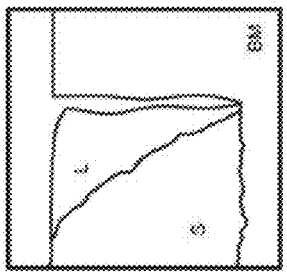




t_3
Vapor pressure forces the keyhole open expelling spatter
Penetration reduced



t_2
Surface tension and gravity attempt to close the keyhole



t_1
Deep penetration with narrow keyhole

(PRIOR ART)

FIG. 1

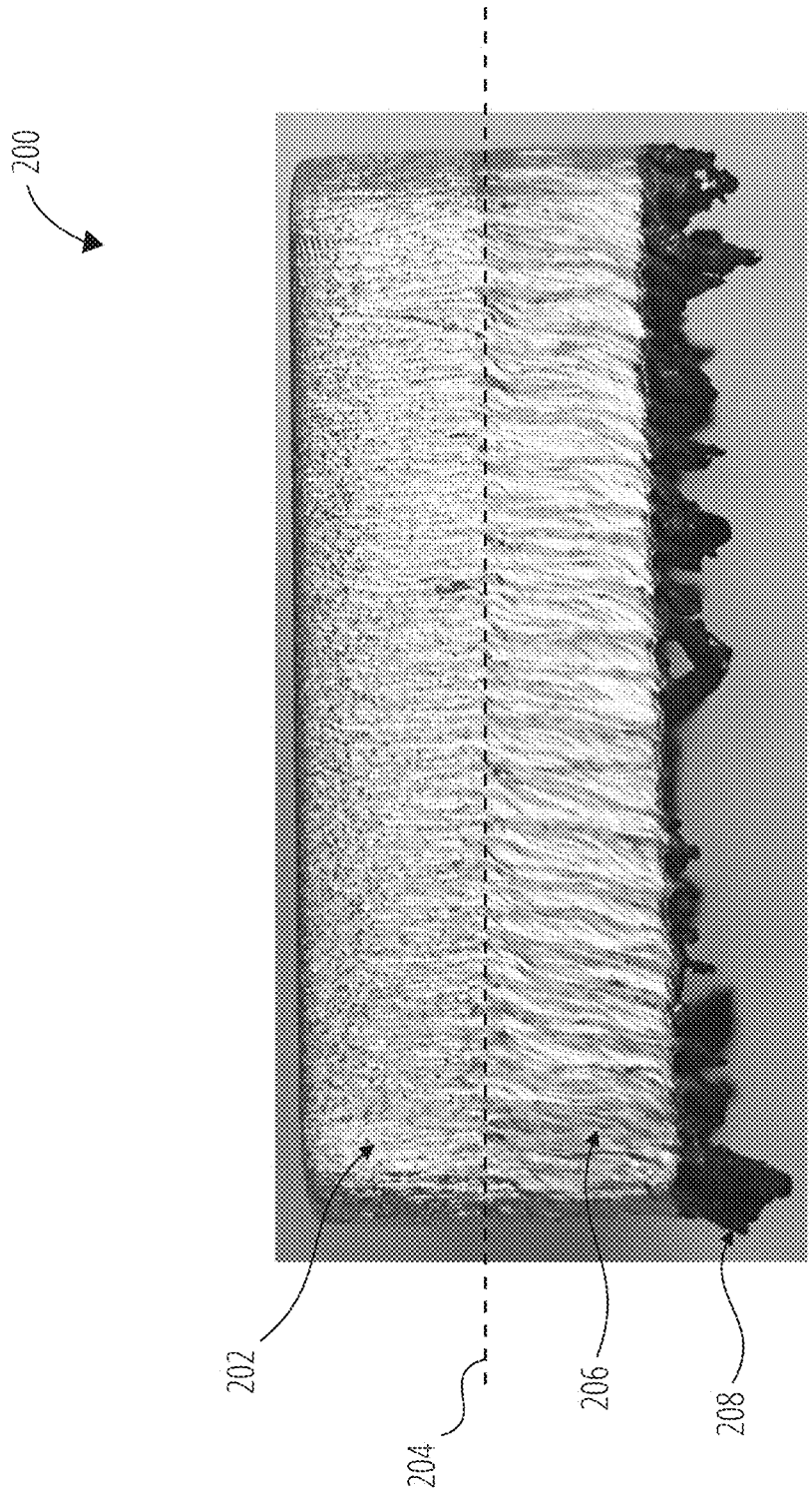


FIG. 2

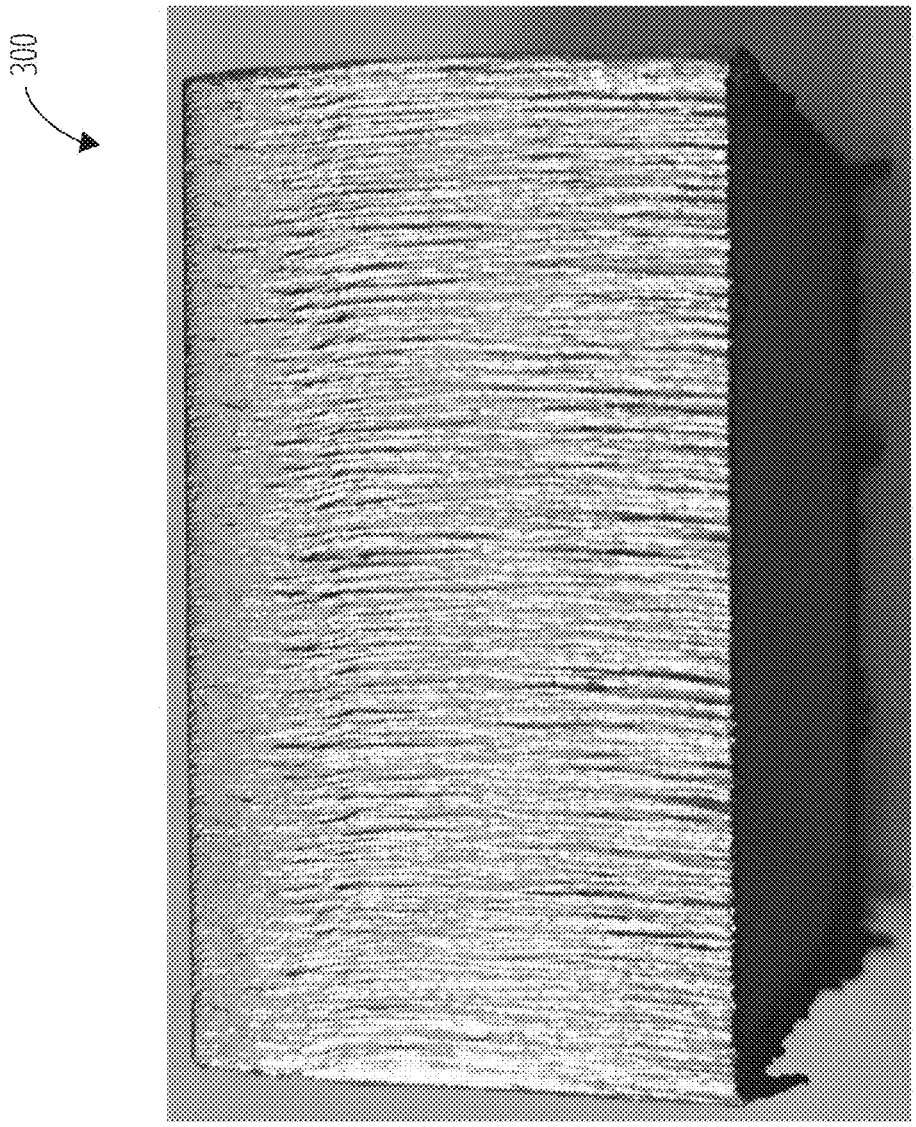


FIG. 3

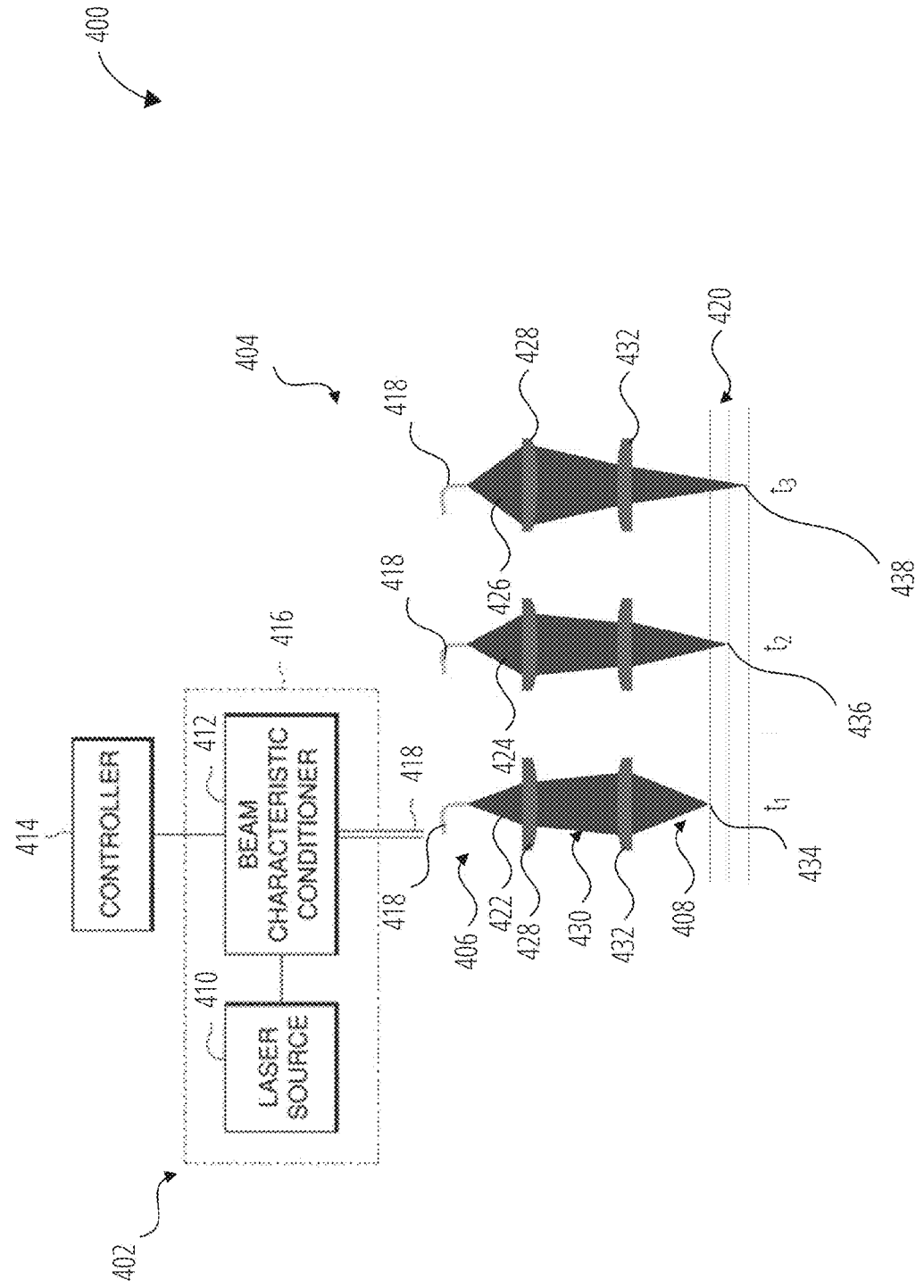

400
402
404
406
408
410
LASER SOURCE
412
BEAM CHARACTERISTIC CONDITIONER
414
CONTROLLER
416
418
420
422
424
426
428
430
432
434
436
438
t_1
t_2
t_3

FIG. 4

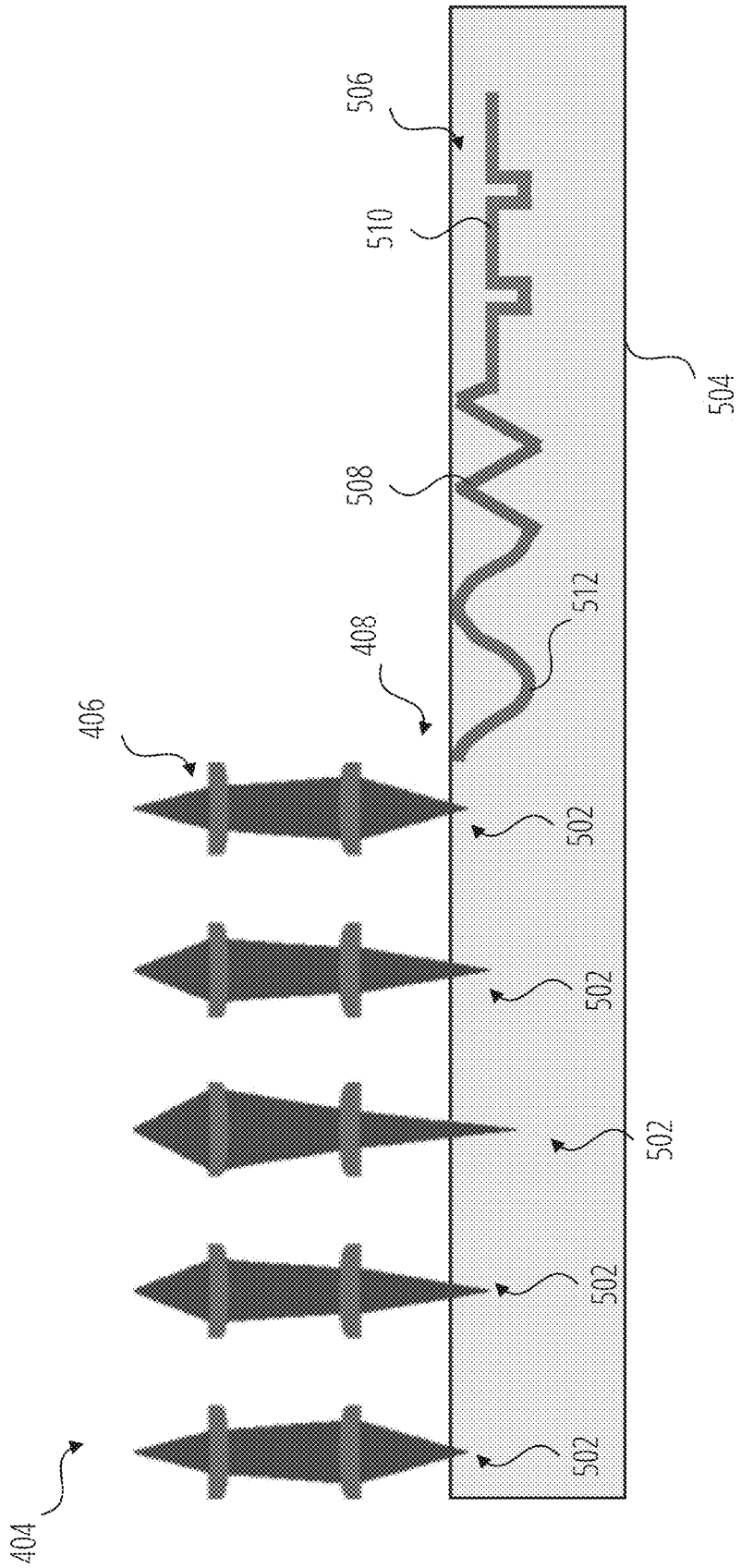


FIG. 5

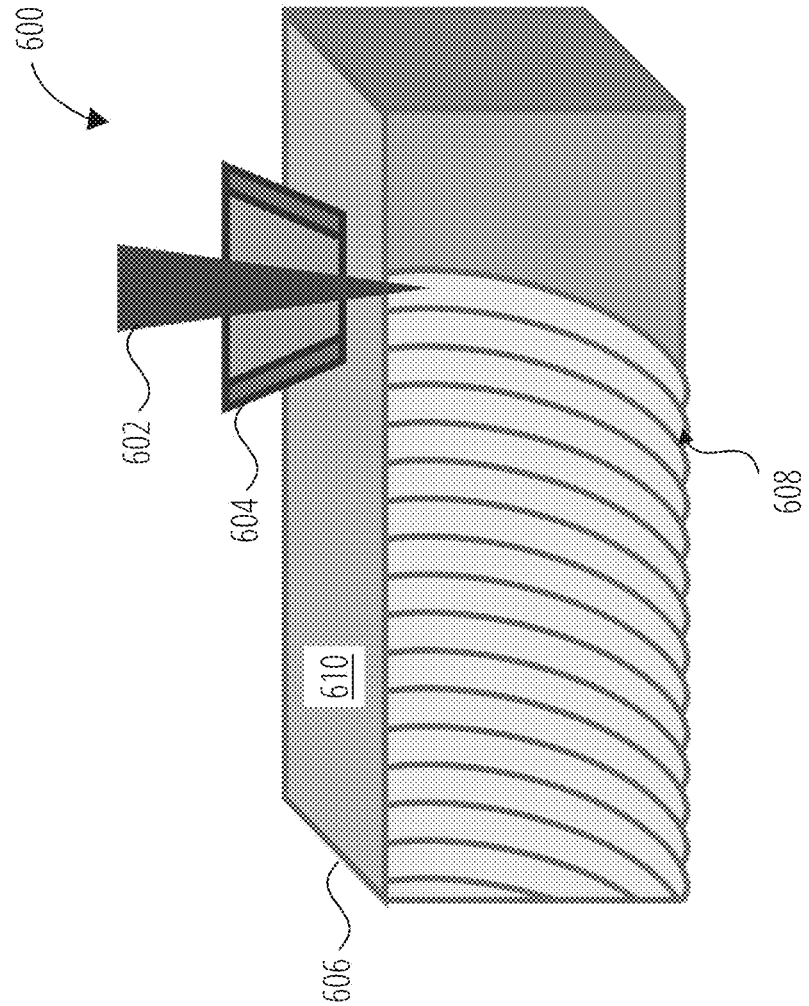


FIG. 6

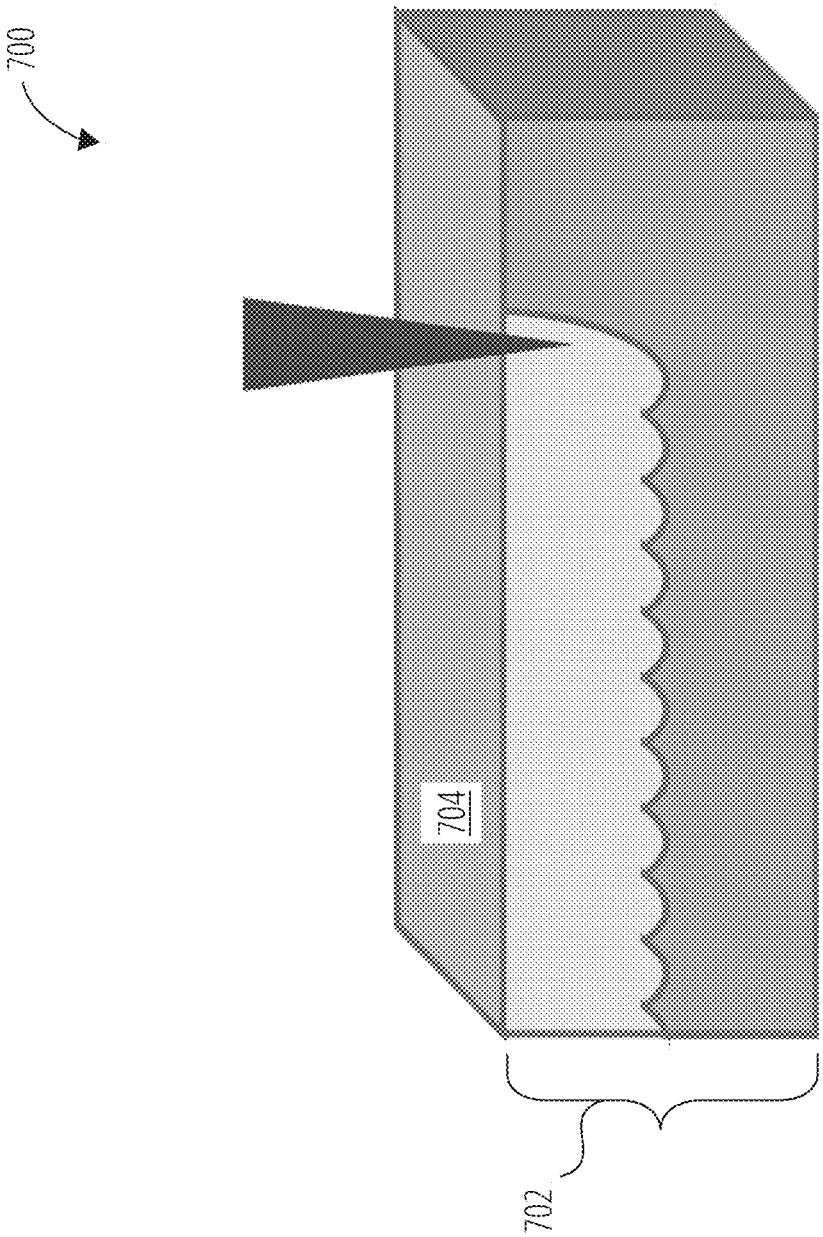


FIG. 7

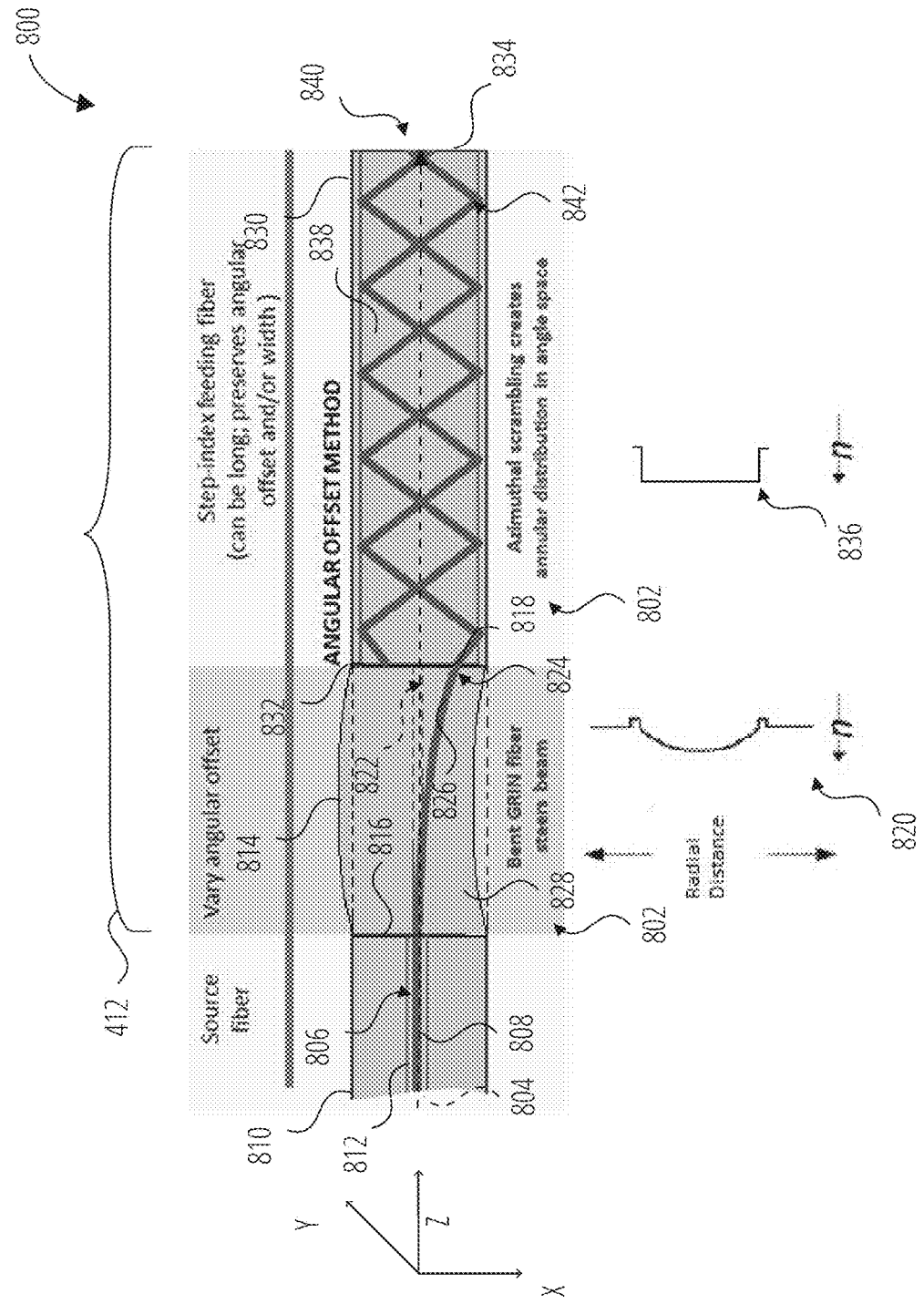

800
412
Source fiber
806
Vary angular offset
814
832
Step-index feeding fiber
(can be long; preserves angular offset and/or width)
830
840
834
838
842
ANGULAR OFFSET METHOD
Azimuthal scrambling creates annular distribution in angle space
818
802
836
n
816
822
826
824
Beat GRIN fiber exits beam
828
802
Radial Distance
n
820
810
812
804
808
Y
Z
X

FIG. 8

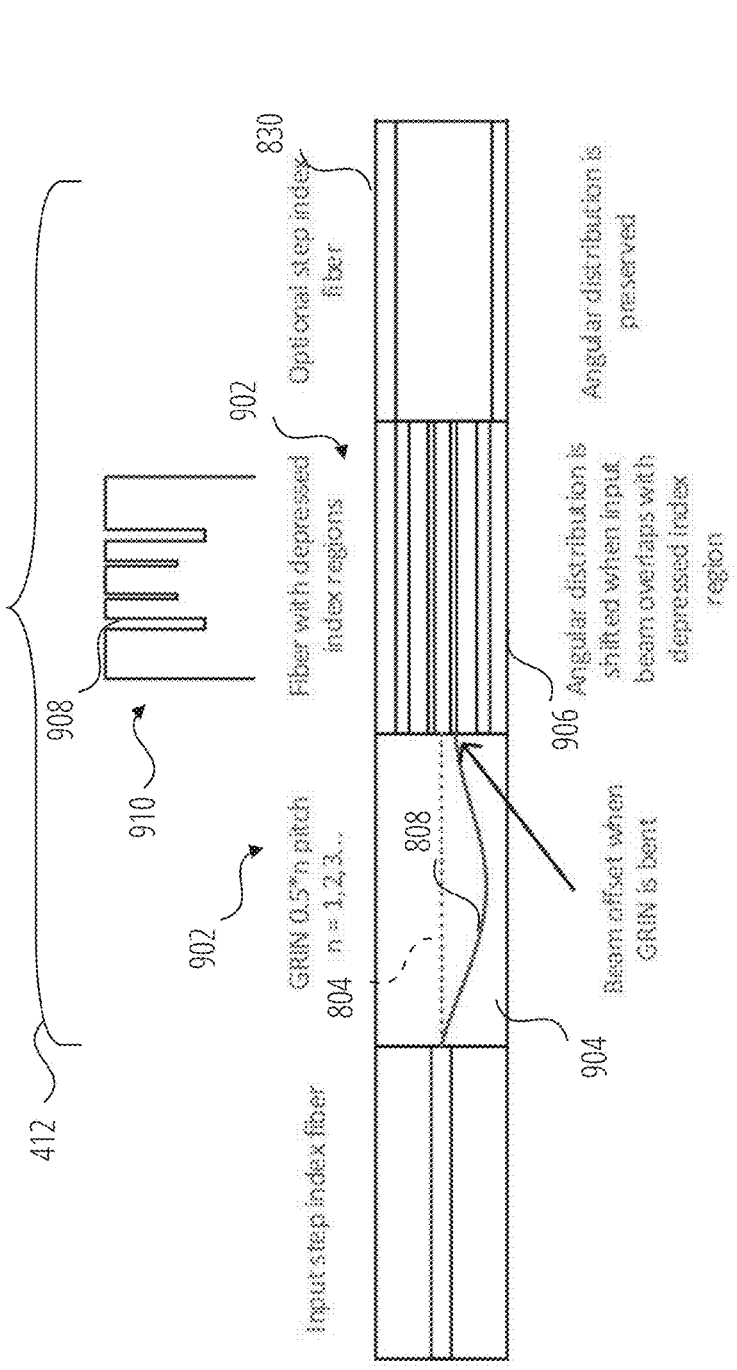

900
412
908
910
902
902
804
808
906
904
830
Input step index fiber
GRIN 0.5*n pitch
n = 1,2,3...
Fiber with depressed index regions
Optional step index fiber
Beam offset when GRIN is bent
Angular distribution is shifted when input beam overlaps with depressed index region
Angular distribution is preserved

FIG. 9

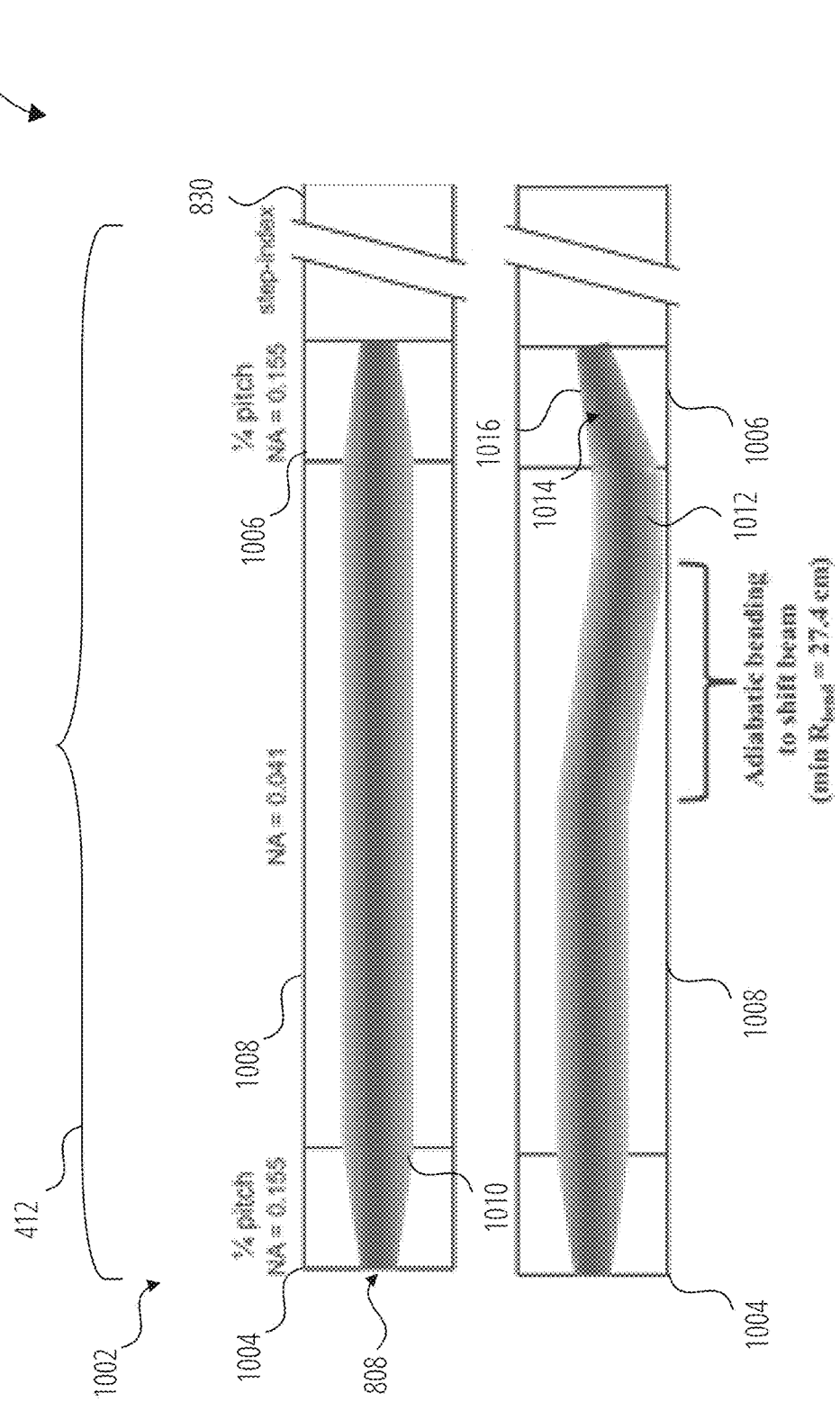

1000
412
1002
830
step-index
¾ pitch
NA = 0.155
1006
1008
NA = 0.041
¾ pitch
NA = 0.155
1004
808
1016
1014
1012
1006
1010
Adiabatic bending
to shift beam
(min R_{bend} = 27.4 cm)
1008
1004

FIG. 10

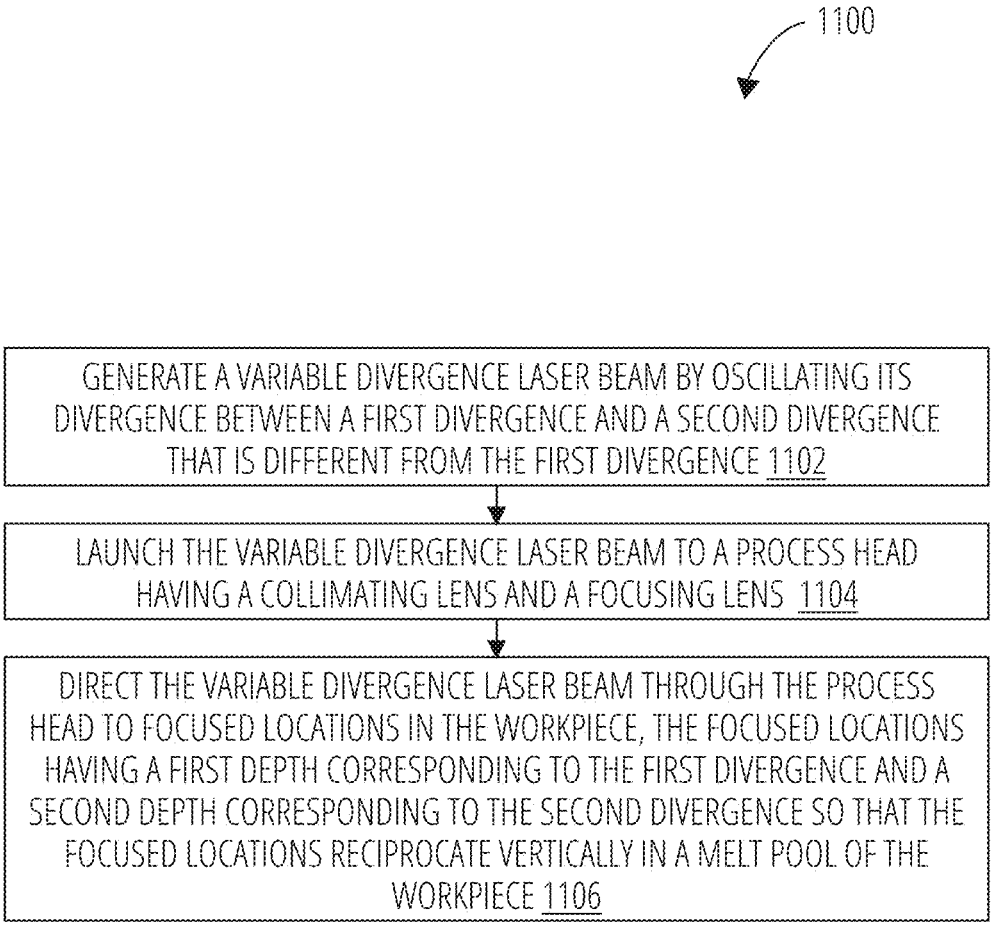

1100
GENERATE A VARIABLE DIVERGENCE LASER BEAM BY OSCILLATING ITS DIVERGENCE BETWEEN A FIRST DIVERGENCE AND A SECOND DIVERGENCE THAT IS DIFFERENT FROM THE FIRST DIVERGENCE 1102
LAUNCH THE VARIABLE DIVERGENCE LASER BEAM TO A PROCESS HEAD HAVING A COLLIMATING LENS AND A FOCUSING LENS 1104
DIRECT THE VARIABLE DIVERGENCE LASER BEAM THROUGH THE PROCESS HEAD TO FOCUSED LOCATIONS IN THE WORKPIECE, THE FOCUSED LOCATIONS HAVING A FIRST DEPTH CORRESPONDING TO THE FIRST DIVERGENCE AND A SECOND DEPTH CORRESPONDING TO THE SECOND DIVERGENCE SO THAT THE FOCUSED LOCATIONS RECIPROCATE VERTICALLY IN A MELT POOL OF THE WORKPIECE 1106

FIG. 11

VARIABLE DIVERGENCE LASER FOR DYNAMIC FOCUS ADJUSTMENT

RELATED APPLICATIONS

[0001] This US Non-Provisional patent application relies for priority on U.S. Provisional Patent Application Ser. No. 63/504,151, entitled "VARIABLE DIVERGENCE LASER FOR DYNAMIC FOCUS ADJUSTMENT," filed May 24, 2023, the content of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates generally to fiber lasers and fiber-coupled lasers. In particular, this disclosure relates to adjusting beam characteristics with an optical fiber for generating a variable beam that is used in material processing.

BACKGROUND INFORMATION

[0003] The applicant for the present application, nLIGHT, Inc. of Camas, Washington, has developed all-fiber solutions for providing variable beam characteristics. These fiber structures are employed in various high-power laser processing applications (e.g., deposition, welding, cutting, scribing, additive manufacturing, or other applications), some of which are the subject of the following nLIGHT patents describing technology developed by Brian M. Victor: U.S. Pat. Nos. 10,668,567; 10,646,963; 10,739,621; and 10,656,330. Some portions of these patents are summarized as follows. For instance, with reference to FIG. 29 of the '567 patent, a multi-operation optical beam delivery device facilitates different laser process operations by modification of beam characteristics. FIGS. 30-31C of the '963 patent and related passages describe techniques for changing beam characteristics of a beam delivery device so as to control a melt pool property (e.g., a signature or a trait) or a keyhole cavity property. FIGS. 31-33 of the '621 patent and related passages describe a laser system for brazing or welding, in which the beam characteristics are modified to accommodate different processes. FIGS. 30-34B of the '330 patent and related passages describe how the beam characteristics can be modified for controlling solidification.

[0004] Some attempts at controlling a beam using free-space optics have been made for metal cutting applications. For example, attempts at cutting of a material have entailed rapidly moving a lens up or down inside a cutting head, using a free-space optic assembly that converts the beam into multiple static coaxial foci, and using a phased array solution to achieve a dynamic focus effect from multiple laser sources.

SUMMARY OF THE DISCLOSURE

[0005] Disclosed is a laser processing system for laser processing a workpiece (e.g., a metal component or other material such as glass, ceramic, or polymer material). In the system, a fiber assembly includes a laser source and a divergence-tuning beam characteristic conditioner configured to, in response to a control input, repetitively change a variable divergence laser beam between a first divergence and a second divergence that is different from the first divergence. A process head has a collimating lens and a focusing lens. A delivery fiber is coupled to the divergence-tuning beam characteristic conditioner for guiding the variable divergence laser beam and launching it to the process head, in which the collimating lens is configured to receive the variable divergence laser beam and direct it as an intermediate beam to the focusing lens. The focusing lens is configured to focus the intermediate beam toward the workpiece at a first depth corresponding to the first divergence and at a second depth corresponding to the second divergence so that a focused location reciprocates vertically in a melt pool of the workpiece.

[0006] The laser processing system may also include the divergence-tuning beam characteristic conditioner being configured to repetitively change the variable divergence laser beam by oscillating between the first and second divergences.

[0007] The laser processing system may also include the divergence-tuning beam characteristic conditioner being configured to repetitively change the variable divergence laser beam by discrete switching between the first and second divergences.

[0008] The laser processing system may also include the divergence-tuning beam characteristic conditioner being further configured to repetitively change the variable divergence laser beam between the first divergence and the second divergence utilizing a constant beam power between the first divergence and the second divergence.

[0009] The laser processing system may also include the divergence-tuning beam characteristic conditioner being further configured to repeatedly change the variable divergence laser beam between the first divergence and the second divergence by, selectively, using one or more frequencies in a first frequency range for welding metal or in a second frequency range for cutting metal. The laser processing system may also include the one or more frequencies of the second frequency range being at higher frequencies than the one or more frequencies of the first frequency range. The laser processing system may also the first frequency range including frequencies of approximately 10 Hertz to 1,000 Hertz, and the second frequency range including frequencies of approximately 1,000 Hertz to 5,000 Hertz.

[0010] The laser processing system may also include the divergence-tuning beam characteristic conditioner having a first and second length of fiber having different RIPs.

[0011] The laser processing system may also include a mandrel for perturbing the divergence-tuning beam characteristic conditioner. The laser processing system may also include a micro-bend section for perturbing the divergence-tuning beam characteristic conditioner.

[0012] The disclosed techniques provide the following advantages. A time averaged beam waist is deeper/longer than a static beam. Varying the location of the focus enables deeper penetration welding with a small spot size. It also improves keyhole stability during vaporization welding. Thicker metal cutting is capable without decreased speed for fusion cutting thick plates (N_2, air, Argon, etc.). There is improved edge quality for cutting thick plate with either fusion cutting (N_2, air, Argon, etc.) or exothermic cutting (O_2). The techniques provide for altering working distance from simple, fixed optics at high speeds for any process. The disclosed techniques are believed to be faster and less expensive than mechanics needed to accelerate and move lens(es) within the head at similar speeds.

[0013] Other technical features may be readily apparent to one skilled in the art from the following figures, description,

and claims. The foregoing and other objects, features, and advantages will become more apparent from the following detailed description, which proceeds with reference to the accompanying figures, which may not be drawn to scale.

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced.

[0015] FIG. **1** is an annotated sequence of longitudinal cross sections during a conventional laser welding process, according to the prior art.

[0016] FIG. **2** is an image of a metallic material after a conventional laser cutting process utilizing a static focus of the laser beam.

[0017] FIG. **3** is an image of another metallic material after a laser cutting process having a dynamic focus adjustment of a laser beam, in accordance with examples described herein.

[0018] FIG. **4** is a block diagram of a laser processing system for generating variable divergence laser beam provided to a process head that produces a dynamic focus adjustment, in accordance with examples described herein.

[0019] FIG. **5** is a time dependent graph of displacement with dwell at different depths/locations in the material for each cutting and welding.

[0020] FIG. **6** is an isometric view of a workpiece being cut during a laser process, according to one embodiment.

[0021] FIG. **7** is an isometric view of a workpiece being welded during a laser process, according to one embodiment.

[0022] FIG. **8** is a side view of an optical beam delivery device including a bendable ¼ pitch length of GRIN fiber, a step-index length of fiber, and a straight GRIN length of fiber, according to one embodiment.

[0023] FIG. **9** is a side view of an optical beam delivery device including multiple lengths of optical fiber, according to another embodiment.

[0024] FIG. **10** is a side view of an optical beam delivery device including multiple lengths of optical fiber, according to another embodiment.

[0025] FIG. **11** is a flowchart of a process, according to one embodiment

DETAILED DESCRIPTION OF EMBODIMENTS

Introduction

[0026] As used in this application and in the claims, the singular forms "a," "an," and "the" include the plural forms unless the context clearly dictates otherwise. Additionally, the term "includes" means "comprises." Further, the term "coupled" does not exclude the presence of intermediate elements between the coupled items.

[0027] The systems, apparatuses, and methods described herein should not be construed as limiting in any way. Instead, the present disclosure is directed toward all novel and non-obvious features and aspects of the various disclosed embodiments, alone and in various combinations and sub-combinations with one another. The disclosed systems, methods, and apparatuses are not limited to any specific aspect or feature or combinations thereof, nor do the disclosed systems, methods, and apparatuses require that any one or more specific advantages be present, or problems be solved. Any theories of operation are to facilitate explanation, but the disclosed systems, methods, and apparatuses are not limited to such theories of operation.

[0028] Although the operations of some of the disclosed methods are described in a particular, sequential order for convenient presentation, it should be understood that this manner of description encompasses rearrangement, unless a particular ordering is required by specific language set forth below. For example, operations described sequentially may in some cases be rearranged or performed concurrently. Moreover, for the sake of simplicity, the attached figures may not show the various ways in which the disclosed systems, methods, and apparatuses can be used in conjunction with other systems, methods, and apparatuses. Additionally, the description sometimes uses terms like "produce" and "provide" to describe the disclosed methods. These terms are high-level abstractions of the actual operations that are performed. The actual operations that correspond to these terms will vary depending on the particular implementation and are readily discernible by skilled persons.

[0029] In some examples, values, procedures, or apparatuses are referred to as "lowest," "best," "minimum," or the like. It will be appreciated that such descriptions are intended to indicate that a selection among many used functional alternatives can be made, and such selections need not be better, smaller, or otherwise preferable to other selections. Examples are described with reference to directions indicated as "above," "below," "upper," "lower," and the like. These terms are used for convenient description, but do not imply any particular spatial orientation. Moreover, in the following examples, laser components and assemblies are described at a high level of abstraction and do not include a complete description of all mechanical, electrical, and optical elements necessary for operation.

[0030] As used herein, optical radiation refers to electromagnetic radiation at wavelengths of between about 200 nm and 10 μm, and typically between about 200 nm and 2 μm. Examples based on available laser diode and fiber laser sources and optical fibers generally are associated with wavelengths of between about 800 nm and 2,000 nm. In some examples, propagating optical radiation is referred to as one or more beams having diameters, asymmetric fast and slow axes, beam cross-sectional areas, and beam divergences that can depend on beam wavelength and the optical systems used for beam shaping. For convenience, optical radiation is referred to as light or beams in some examples and need not be at visible wavelengths. Forward-propagating light or optical beams or beam portions refer to light, beams, or beam portions that propagate in a direction of normal emission. Backward-propagating light or optical beams or beam portions refer to light, beams, or beam portions that propagate in an opposite direction of normal emission.

Component Overview

[0031] To understand the characteristics of embodiments described in the present disclosure, it is helpful to first understand the underlying components and physical phenomena exploited with the components. Accordingly, the following paragraphs summarize various designs and properties of optical fibers and lenses.

[0032] Optical fibers: An optical fiber is a thin, flexible strand of material (typically glass) that guides light within a core that is surrounded by a cladding. The core has a higher refractive index than the cladding, causing light to be guided by total internal reflection. The refractive index profile (RIP), i.e., the refractive index as a function of position transverse to the fiber axis, determines many of the important properties of the fiber. Many but not all fibers are cylindrically symmetric (i.e., the index is independent of azimuthal angle about the fiber axis). Optical fibers may have circular, square, rectangular, polygonal, oval, elliptical, or other cross-sections. In some embodiments, the core and claddings are approximately concentric. In other examples, one or more of the cores and claddings are decentered, and in some examples, core and cladding orientation and/or displacement vary along a waveguide length.

[0033] Optical fibers are typically formed of silica (glass), and dopants are added to increase or decrease the refractive index. Dopant concentrations are varied transversely to generate the desired RIP. In some examples, fibers or other waveguides are made of other materials such as fluorozirconates, fluoroaluminates, fluoride or phosphate glasses, chalcogenide glasses, or crystalline materials such as sapphire, depending on wavelengths and other properties of interest. Refractive indices of silica and fluoride glasses are typically about 1.5, and refractive indices of other materials such as chalcogenides can be 3 or more. In still other examples, optical fibers can be formed in part or completely of plastics (polymers).

[0034] In some examples, a doped waveguide core such as a fiber core provides optical gain in response to pumping. In the examples disclosed herein, a waveguide core such as an optical fiber core is doped with one or more rare-earth elements such as Nd, Yb, Ho, Er, Tm, or other active dopants or combinations thereof. Such actively doped cores can provide optical gain in response to optical pumping. As disclosed below, waveguides having such active dopants can be used to form optical amplifiers, or, if provided with suitable optical feedback such as reflective layers, mirrors, Bragg gratings, or other feedback mechanisms, such waveguides can generate laser emissions. Optical pump radiation can be arranged to co-propagate and/or counter-propagate in the waveguide with respect to a propagation direction of an emitted laser beam or an amplified beam.

[0035] Step-index fibers: The most common RIP is known as "step index," in which the core has a uniform index and is surrounded by the cladding with a uniform (lower) index. In a step-index fiber, light launched into the fiber core in a given location tends to spread radially and azimuthally to fill the core, i.e., the core does not preserve spatial information about the location of the launched beam. It is known that light rays launched at a particular angle with respect to the fiber axis tend to exit the fiber core at the same angle (in a cone because of azimuthal scrambling), i.e., step-index fibers typically preserve angular information. In particular, the preservation of the launch angle at the output of the step-index fiber (i.e., the far-field divergence profile) is a property leveraged by embodiments of this disclosure. (Note that the terms far- and near-field intensity distribution mean, respectively, an angular intensity distribution (with respect to the optical axis) and a transverse spatial intensity distribution.) Various effects can cause imperfect preservation of the launch angle (e.g., causing higher divergence for the output than the input rays), and multiple studies have investigated methods to minimize this effect.

[0036] GRIN fibers: Some fibers have a non-uniform RIP in the core and (less often) in the cladding. A common, non-uniform core RIP is known as "graded index" or "GRIN," in which the index value decreases monotonically from the center of the core to the cladding. In many cases, the GRIN profile is parabolic. The size of a beam coupled into a parabolic GRIN fiber may oscillate periodically as the beam propagates along the fiber, with the oscillation period known as the "pitch" of the fiber. A parabolic GRIN fiber with an appropriate length can act as a lens that focuses or collimates a beam. This property is also used in free-space GRIN lenses, i.e., it is not limited to fibers.

[0037] Multi-clad fibers: Another example of a non-uniform RIP is a fiber with multiple claddings ("double-clad fiber," "triple-clad fiber," etc.). As with other non-uniform RIPs, propagation of light in a multi-clad fiber depends on the specific RIP and the launch or coupling conditions. Light launched into a given region will spread into all adjacent higher-index regions, but it may not spread into all adjacent lower-index regions (which act as a cladding for the higher-index region).

[0038] Multi-core fibers: Some fibers have multiple cores, i.e., multiple high-index guiding regions separated by surrounding low-index regions. The cores can have different sizes and shapes. A ring fiber, described above, is an example of a multi-core fiber. Light launched into a given core of a multi-core fiber will typically spread out to fill that core (as in a fiber with a single core) but will not spread into the other cores because the surrounding low-index region (cladding) prevents propagation between the cores. Multi-core fibers can be used to generate different beam shapes (i.e., near-field spatial profiles) by coupling different powers into each core. In comparison to standard fibers (especially step-index fibers), multi-core fibers have several disadvantages: (1) Multi-core fibers are generally more expensive than standard fibers. (2) The low-index regions separating the guiding regions generate highly divergent light if a portion of the beam is coupled into them. The downstream optics are designed to accommodate this excess divergence, which typically adds cost, complexity, and/or optical loss (reduced efficiency). If not properly managed, the excess divergence can cause overheating or damage of the optics or other components. (3) The dimensions of the guiding regions are fixed when the fiber is manufactured. The beam dimensions thus cannot be continuously varied (although the power in each region can be continuously varied). Furthermore, if an application prefers different dimensions (e.g., a larger diameter ring), a new multi-core fiber must be fabricated, which can lead to a proliferation of product designs (which complicates and adds cost to manufacturing). (4) Splicing of multi-core fibers is complicated by the need to align multiple guiding regions, which each have fabrication tolerances. Achieving the desired splicing performance can require excessively tight fiber tolerances (increasing cost), low splice yields, and/or performance degradation upon splicing (e.g., from coupling light into the low-index regions, resulting in increased divergence).

[0039] Note that the above discussion of the different types of optical fibers primarily pertains to multimode fibers, in which the ray picture of light propagation is accurate.

Single-mode or near-single-mode fibers introduce other phenomena that are less relevant to the embodiments described in this disclosure.

[0040] Lenses: Lenses are ubiquitous in optical systems. A lens is conventionally thought of as a piece of glass with one or both surfaces curved, and the angle of propagation of a light ray is influenced by refraction at the surfaces. Other implementations of lenses are possible, including a GRIN lens (as mentioned above) in which the RIP of the material varies with radial position (rather than employing curved surfaces). The end of a fiber or of an end cap attached to a fiber can have a curved surface to function as a lens. A design property of a lens is its focal length. A collimated input beam is focused to a point one focal length away from the lens (in the ray picture). Similarly, a point source located one focal length away from a lens is collimated at the lens output. Stated more generally, a lens maps between position and angle, i.e., a lens converts the angle of a ray at the input focal plane to position at the output focal plane, and it converts the input position of a ray at the input focal plane to angle at the output focal plane.

[0041] FIG. **1** shows an example of keyhole instability fluctuations in a conventional chaotic laser welding process. The weld is formed by melting parts of base metal with optional filler material. A high-intensity beam (not shown) vaporizes liquid metal at the bottom of a cavity, called a keyhole. As the vapor escapes, creating this keyhole, the depression remains narrow due to its liquid walls which are constantly attempting to close due to gravity and surface tension. Any fluctuation in the material or the process conditions can cause the keyhole to momentarily collapse. The encroaching liquid metal prevents the laser from reaching the solid material. Consequently, the laser cannot effectively continue welding the deeper material. The keyhole can then restart, for instance, at t**4** (not shown), reestablishing penetration without keyhole occlusion. However, the keyhole may collapse once more. This dynamic instability during the process is undesirable, as it affects penetration and leads to spatter (i.e., ejected metal particles).

[0042] To avoid this chaotic instability, the focus position can be dynamically adjusted to maintain a minimal disturbance of the liquid walls of the keyhole. Surface tension effects of the keyhole walls can be controlled by keeping the geometry of the keyhole in a dynamic state. When the keyhole is opening or closing the rate of change prevents the liquid metal from chaotically trying to collapse. By dynamically adjusting the focus position, the keyhole can be designed to minimally increase and decrease in size and depth. This dynamic adjustment of the keyhole controls the liquid metal flow and prevents the liquid material from reaching a steady state of chaotic instability.

[0043] FIG. **2** shows a metal workpiece material **200** after a conventional laser cutting process. In this example, metal workpiece material **200** is about 12 mm thick and has been cut using a 4 kW laser beam having a static focus. The cutting speed was about 0.5-0.8 m/min with focus of −8 mm into the material (measured from the top surface). In general, the focus position for inert cutting is typically beyond half-way through the thickness of the material or near the surface of the material. The focus position for exothermic cutting is typically at the surface or above the material surface.

[0044] An inefficient cutting process occurs when the cut kerf is unstable or too narrow preventing clean ejection of the molten metal during cutting. As a result, an upper portion **202** (i.e., the cut portion above a mid-thickness line **204**) is relatively smooth whereas a lower portion **206** is relatively rough. A bottom side includes a significant amount of dross **208**.

[0045] The present inventor recognized, however, that if the focus location of the beam is reciprocated within the cut kerf, then it is possible to widen the kerf and more easily eject the liquid metal during cutting. The wider diameters of the beam at the extents of the waist provide more melting of the sidewalls of the kerf. This additional material removal can help open the cut kerf to more easily allow ejection by the vaporization and pressurized assist gas used in cutting process. To avoid overheating the material, oscillating the position of the focus within the material thickness reduces the interaction time at each position along the cut front.

[0046] In contrast to FIG. **2**, FIG. **3** shows a metal workpiece material **300** subjected to an improved cutting process. This process uses a dynamic focus adjustment of a laser beam in which the focus is rapidly moved vertically through the thickness of the material undergoing the cut. In this example, metal workpiece material **300** is about 16 mm, the profile of the cut is relatively smooth throughout the thickness, and there is less dross at a cutting speed of 0.4-0.8 m/min. By dynamically varying or oscillating the focus plane during processing improves the cutting parameters such as edge quality, reduction in dross, the smoothness of the edge, roughness, and linear speed.

[0047] FIG. **4** shows a laser processing system **400**, which includes an optical beam delivery device **402** and a process head **404**. Optical beam delivery device **402** generates a variable divergence laser beam **406** that, when coupled to process head **404**, provides the dynamic focus adjustment of an output laser beam **408**, in accordance with examples described herein.

[0048] Optical beam delivery device **402** includes a laser source **410**, a beam characteristic conditioner **412**, and a controller **414**. As described in the '567 patent, laser source **410** and beam characteristic conditioner **412** are components of a fiber assembly **416**. Laser source **410**, which may be a step-index feeding fiber or other type of optical fiber, provides an input laser beam (not shown) that is coupled to beam characteristic conditioner **412**. Beam characteristic conditioner **412** includes first and second lengths of fiber operable to provide variable divergence laser beam **406**. Examples of beam characteristic conditioner **412** are shown and described later with reference to FIG. **8**-FIG. **10**.

[0049] Controller **414** facilitates a selection of an amount of perturbation applied to beam characteristic conditioner **412** so as to vary the divergence of variable divergence laser beam **406** that is guided to process head **404**. Various types of perturbation devices are shown in FIG. 24 of the '597 patent and include micro-bends, acoustic-optic transducer, and piezo electric device. Divergence structures (configured to modify divergence in response to perturbation) are shown in FIGS. 14B and 20 of the '597 patent.

[0050] A delivery fiber **418** from beam characteristic conditioner **412** may be routed directly to process head **404** or workpiece **420**. In other embodiments, delivery fiber **418** may be routed to a fiber-to-fiber coupler (FFC) or fiber-to-fiber switch (FFS), which couples the light from delivery fiber **418** into a process fiber (not shown) that transmits the beam to process head **404** or the workpiece.

[0051] Process head **404** is shown operating at three different times: t_1, t_2, and t_3, which correspond to three different states of divergence of variable divergence laser beam **406**. For instance, at time t_1, variable divergence laser beam **406** has a narrowest divergence **422** corresponding to a first amount of perturbation applied to beam characteristic conditioner **412**. At time t_2, variable divergence laser beam **406** has an intermediate divergence **424** corresponding to a second amount of perturbation applied to beam characteristic conditioner **412**. And at time t_3, variable divergence laser beam **406** has a widest divergence **426** corresponding to a third amount of perturbation applied to beam characteristic conditioner **412**.

[0052] As variable divergence laser beam **406** is applied to a collimating lens **428** at the non-deal distance, variable divergence laser beam **406** is directed as an intermediate beam **430** to a focusing lens **432**. Intermediate beam **430**, however, has different angular and intensity distributions at a focusing lens **432** depending on the input divergence from delivery fiber **418** and the offset of collimating lens **428** from the ideal collimation distance. Accordingly, as focusing lens **432** focuses intermediate beam **430** at time t_1, t_2, and t_3, a focal point is moved down (or up) the beam axis for focusing at, respectively, a first material depth **434** (e.g., atop an upper surface of a workpiece **420** shown as a cross-section), a second material depth **436** (e.g., at an intermediate depth in the workpiece), and a third material depth **438** (e.g., at a lowest point in melt pool of the workpiece). The depths of the output laser beam **408** may be discrete depths or output laser beam **408** may be continuously transitioned between the depths. Furthermore, the depths may be reciprocated (e.g., rapidly adjusted) between shallow and deep depths as output laser beam **408** translates laterally relative to the workpiece. As output laser beam **408** is cutting workpiece **420**, the focus is moving up and down rapidly such that the highest intensity is at the focus beam, and that intensity is moved throughout the material.

[0053] FIG. **5** shows on the right and left sides, respectively, examples how process head **404** has its focus positions **502** adjusted vertically over time as it moves relative to a workpiece **504**, and examples of different resulting process paths **506** in or on workpiece **504**. For instance, the frequency of focus position **502** can be adjusted in the range of 1 Hz to 10 kHz. When this vertically oscillating focus position **502** is combined with the travel speed of the cutting or welding process, in the range of 0.1 m/min to 500 m/min, the focus depth of the beam over the translation of the material is designed to match any repeating or arbitrary waveform used to optimize the process. This synchronization of focus and process speed results in a sawtooth **508** or triangle waveform for keyhole welding to maintain constant opening or closing condition of the keyhole vapor cavity. For cutting, this waveform can be a square wave **510** alternating between high and low focus to optimize the cut front and kerf width in the material. Other shapes, such as a sinusoidal shape **512**, are also possible. In this example, the wavelength is on the order of microns to millimeters.

[0054] FIG. **6** shows an example of a laser cutting process **600**. A laser beam **602** melts and vaporizes some of the base material in the beam path, while compressed gas delivered from a coaxial nozzle **604** ejects the melted material from the cut kerf. As described previously, laser beam **602** has the location of its intensity vertically reciprocated as laser beam **602** moves laterally to cut through a metal workpiece **606**. The oscillating laser leaves striations on a cut edge **608** with a pattern caused by the processing variables. The vertical displacement of laser beam **602** relative to a top surface **610** of workpiece **606** is in a range of +30 mm to −50 mm so as to optimize material ejection, edge finish, and speed by regulating the dynamics of the absorption, melting, and ejection of the material from the kerf. In some embodiments, workpiece **606** has a thickness ranging from 3 mm to 100 mm. The frequency of the vertical reciprocation is in the thousands of Hz, such as 2,000 Hz. The cutting speed is typically ranging from 1 meter/min to 500 meters/min, in some embodiments.

[0055] FIG. **7** shows an example of a laser welding process **700** that is partially penetrating the full material thickness **702**. In this example, the vertical displacement relative to a top surface **704** is in a range of +10 mm to −10 mm to regulate the liquid/gas dynamics of the keyhole vapor cavity. The thickness of the material can vary, for example, from 3 mm to 30 mm. The variable focus welding process can be designed for complete joint penetration (not shown) or partial penetration welding. The frequency of the vertical focus reciprocation is in the hundreds of Hz (e.g., 10 Hz to 1,000 Hz). The welding speed would typically range from 1 meter/min to 30 meters/min, in some embodiments.

[0056] FIG. **8** shows an optical beam delivery device **800**, which includes an embodiment of beam characteristic conditioner **412**. In this example, optical beam delivery device **800** includes multiple lengths of optical fiber **802** arranged along an optical axis **804** and configured to generate variable divergence laser beam **406** by modifying a divergence distribution **806** of an input beam **808**.

[0057] In one embodiment of FIG. **8**, input beam **808** is delivered from a source fiber **810** having a step-index core **812**. In other embodiments, a source fiber is a ring fiber (e.g., one or more annular cores surrounding an optional central core), and the ring fiber may be a component of beam characteristic conditioner **412** as described in the '597 patent. The term source fiber is any fiber (feeding or process) providing an input beam and that is fiber coupled (e.g., spliced) to a beam characteristic conditioner **412**. The term fiber coupled (or couplable) includes a direct physical connection as well as connections that are slightly spaced apart, i.e., at or within one Rayleigh range, which is the distance along the propagation direction of a beam from the waist to the place where the area of the cross section is doubled.

[0058] Source fiber **810** is spliced to a first length of optical fiber **814**, which has a first input **816** and a first output **818**. Thus, first input **816** is coupled to a source fiber **810** and is configured to receive therefrom input beam **808** that is azimuthally symmetric with respect to optical axis **804**.

[0059] First length of optical fiber **814** has a first RIP **820**. First RIP **820** is configured to produce, in response to a controllable perturbation (e.g., macro-bending), a change in divergence distribution **806** from a first divergence distribution **822** corresponding to input beam **808** to a second divergence distribution **824** corresponding to a modified beam **826** at first output **818**.

[0060] In the example of FIG. **8**, first length of optical fiber **814** is a relatively short (e.g., less than 10 mm) GRIN fiber having a ¼-pitch length. At this length, input beam **808** beam reaches its maximum diameter, which is less than a 300 μm diameter GRIN core **828** in some embodiments. In other embodiments, an initial GRIN optical fiber segment

may have a length of about ¼-pitch+N*½ pitch, where N equals zero or any positive integer. Other lengths are also possible, depending on the desired application (see, e.g., FIG. **10**). In a GRIN fiber, light rays transmit along sinusoidal paths. The length of fiber corresponding to one sine wave period is called the pitch, defined as follows:

$$p = \frac{2\pi}{\sqrt{k}},$$

[0061] where k is the gradient constant determined by the refractive index profile of the GRIN fiber. The refractive index profile of the GRIN fiber is defined as set forth in W. J. Smith, Modern Optical Engineering, Third Edition, McGraw-Hill, Inc. (2000), p. 286:

$$n(r) = n_0\left(1 - \left(\frac{k}{2}\right)r^2\right),$$

[0062] where n_0 is the axial refractive index of the GRIN fiber.

[0063] In some embodiments, first length of optical fiber **814** and an initial section of a second length of optical fiber **830** are bent a variable amount (or not at all) around a mandrel (not shown) or other perturbation device shown and described in FIG. 24 of the '597 patent. Divergence distribution **806** of input beam **808** beam is thereby varied by changing its angular offset without necessarily affecting its width. For instance, first divergence distribution **822** has negligible angular offset with respect to optical axis **804** when first length of optical fiber **814** is not perturbed (i.e., straight, as indicated by dashed lines on first length of optical fiber **814**) whereas second divergence distribution **824** includes a relatively large angular offset when first length of optical fiber **814** is perturbed (e.g., bent, as indicated in FIG. **8**).

[0064] Second length of optical fiber **830**, which includes a second input **832** and a second output **834**, is coupled to first output **818** and configured to receive therefrom modified beam **826**. Second length of optical fiber **830** has a step-index RIP **836** configured to preserve second divergence distribution **824**. Modified beam **826** enters a step-index fiber core **838** (e.g., core diameter of 300 μm) and propagates while substantially preserving the angular offset with respect to optical axis **804**. Second divergence distribution **824** is preserved, i.e., its ray angles relative to optical axis **804** (i.e., the fiber core) remain substantially constant from second input **832** to second output **834**. Although it is largely preserved, for purposes of clarity (and recognizing that there may be some minor variation in the distributions) this disclosure refers to second divergence distribution **824** at second output **834** as a preserved angular intensity distribution **840**.

[0065] In some embodiments, second length of optical fiber **830** may be about five meters to about 50 meters long so as to transport modified beam **826** a significant distance from the source (e.g., to a processing head or tool). As modified beam **826** travels through second length of optical fiber **830**, second divergence distribution **824** is azimuthally averaged (i.e., smoothed or symmetrized) due to normal routing (coiling or bending) of this delivery fiber. Thus, phi ray angles (azimuthal) are assumed to be scrambled in second length of optical fiber **830**. The angular width of input beam **808** results from rays having slightly different angles. These rays take different paths (generally helical) and end up being scrambled azimuthally (cylindrically symmetric, phi invariant).

[0066] In another embodiment, FIG. **9** shows an optical beam delivery device **900**, which includes an embodiment of beam characteristic conditioner **412**. A first length of optical fiber **902** includes a first portion **904** and a second portion **906** coupled to first portion **904**. First portion **904** includes a GRIN optical fiber segment having a length of about N*½ pitch, where N is any positive integer. Other lengths are also possible, depending on the design, but the length of about N*½ pitch facilitates interaction of the beam with low-index divergence structures **908** in second portion **906**. Divergence structures **908** (shown in RIP **910**) are designed to increase divergence. When first portion **904** is perturbed, input beam **808** is displaced from optical axis **804** so that it is incident upon at least one of divergence structures **908**, which increases divergence to provide a modified beam to second length of optical fiber **830**.

[0067] FIG. **10** shows another embodiment of an optical beam delivery device **1000**, which includes another embodiment of beam characteristic conditioner **412**. A first length of fiber **1002** has an input GRIN portion **1004**, an output GRIN portion **1006**, and a central GRIN portion **1008** therebetween. Input GRIN portion **1004** is configured to collimate input beam **808** to provide a collimated beam **1010**. Central GRIN portion **1008** is configured to shift collimated beam **1010** in response to controllable perturbation (e.g., bending, as shown in bottom diagram of FIG. **10**) so as to provide a shifted beam **1012**. Output GRIN portion **1006** is configured to focus shifted beam **1012** to provide modified beam **826** having an angular offset **1014** and produce a divergence distribution that is different from that of input beam **808**. For instance, by employing different NAs for input GRIN portion **1004** and output GRIN portion **1006**, the angular width (divergence) may also be tuned. Input GRIN portion **1004** and output GRIN portion **1006** are each shown as having ¼-pitch length, but other lengths are also possible. The effective focal length of these segments may be designed to provide different divergences of modified beam **1016** or to change other beam properties to accommodate the desired use case. In contrast to ¼- or ½-pitch lengths described previously for first length of optical fiber **814** (FIG. **8**), optical beam delivery device **1000** also provides an example of how first length of fiber **1002** may have varied lengths for different applications.

[0068] FIG. **11** shows a process **1100** for material processing a workpiece with a laser. In block **1102**, process **1100** generates a variable divergence laser beam by oscillating its divergence between a first divergence and a second divergence that is different from the first divergence. In block **1104**, process **1100** launches the variable divergence laser beam to a process head having a collimating lens and a focusing lens. In block **1106**, process **1100** directs the variable divergence laser beam through the process head to focused locations in the workpiece, the focused locations having a first depth corresponding to the first divergence and a second depth corresponding to the second divergence so that the focused locations reciprocate vertically in a melt pool of the workpiece.

[0069] Process **1100** may further includes perturbing a divergence-tuning beam characteristic conditioner for gen-

erating the variable divergence laser beam. Process **1100** may also include micro-bending or macro-bending.

[0070] Process **1100** may also include the divergence-tuning beam characteristic conditioner having first and second lengths of fiber with different RIPs. A first RIP of the first length of fiber is configured to cause a change in a divergence distribution that is preserved by the second fiber having a second RIP.

[0071] Process **1100** may also include changing a frequency of the oscillating based on material type of the workpiece (e.g., metal, glass, ceramic, plastic, or other type), based on thickness of the workpiece, or based on a type of the laser process (e.g., cutting, welding, or other process).

[0072] Having described and illustrated the general principles of examples of the presently disclosed technology, it should be apparent that the examples may be modified in arrangement and detail without departing from such principles. Skilled persons, therefore, will appreciate that many changes may be made to the details of the above-described embodiments without departing from the underlying principles of the invention. The scope of the present invention should, therefore, be determined only by the following claims.

What is claimed is:

1. A laser processing system for material processing a workpiece with a laser, comprising:

a fiber assembly including a laser source and a divergence-tuning beam characteristic conditioner configured to, in response to a control input, repetitively change a variable divergence laser beam between a first divergence and a second divergence that is different from the first divergence;

a process head having a collimating lens and a focusing lens; and

a delivery fiber coupled to the divergence-tuning beam characteristic conditioner for guiding the variable divergence laser beam and launching it to the process head,

in which the collimating lens is configured to receive the variable divergence laser beam and direct it as an intermediate beam to the focusing lens, and in which the focusing lens is configured to focus the intermediate beam toward the workpiece at a first depth corresponding to the first divergence and at a second depth corresponding to the second divergence so that a focused location reciprocates vertically in a melt pool of the workpiece.

2. The laser processing system of claim **1**, in which the divergence-tuning beam characteristic conditioner is configured to repetitively change the variable divergence laser beam by oscillating between the first and second divergences.

3. The laser processing system of claim **1**, in which the divergence-tuning beam characteristic conditioner is configured to repetitively change the variable divergence laser beam by discrete switching between the first and second divergences.

4. The laser processing system of claim **1**, in which the divergence-tuning beam characteristic conditioner is further configured to repetitively change the variable divergence laser beam between the first divergence and the second divergence utilizing a constant beam power between the first divergence and the second divergence.

5. The laser processing system of claim **1**, in which the divergence-tuning beam characteristic conditioner is further configured to repeatedly change the variable divergence laser beam between the first divergence and the second divergence by, selectively, using one or more frequencies in a first frequency range for welding metal or in a second frequency range for cutting metal.

6. The laser processing system of claim **5**, in which the one or more frequencies of the second frequency range are higher frequencies than the one or more frequencies of the first frequency range.

7. The laser processing system of claim **5**, in which:

the first frequency range comprises frequencies of approximately 10 Hertz to 1,000 Hertz; and

the second frequency range comprises frequencies of approximately 1,000 Hertz to 5,000 Hertz.

8. The laser processing system of claim **1**, in which the divergence-tuning beam characteristic conditioner includes a first and second length of fiber having different RIPs.

9. The laser processing system of claim **1**, further comprising a mandrel for perturbing the divergence-tuning beam characteristic conditioner.

10. The laser processing system of claim **1**, further comprising a micro-bend section for perturbing the divergence-tuning beam characteristic conditioner.

11. The laser processing system of claim **1**, in which the material processing is a cutting or welding process.

12. The laser processing system of claim **1**, in which the workpiece is metal.

13. A method for material processing a workpiece with a laser, the method comprises:

generating a variable divergence laser beam by oscillating its divergence between a first divergence and a second divergence that is different from the first divergence;

launching the variable divergence laser beam to a process head having a collimating lens and a focusing lens; and

directing the variable divergence laser beam through the process head to focused locations in the workpiece, the focused locations having a first depth corresponding to the first divergence and a second depth corresponding to the second divergence so that the focused locations reciprocate vertically in a melt pool of the workpiece.

14. The method of claim **13**, further comprising perturbing a divergence-tuning beam characteristic conditioner for generating the variable divergence laser beam.

15. The method of claim **14**, in which the perturbing comprising micro-bending or macro-bending.

16. The method of claim **15**, in which the divergence-tuning beam characteristic conditioner includes first and second lengths of fiber having different RIPs, a first RIP of the first length of fiber causing a change in a divergence distribution that is preserved by the second fiber having a second RIP.

17. The method of claim **13**, further comprising changing a frequency of the oscillating based on material type of the workpiece.

18. The method of claim **13**, further comprising changing a frequency of the oscillating based on thickness of the workpiece.

19. The method of claim **13**, further comprising changing a frequency of the oscillating based on a type of the laser process.

20. The method of claim **13**, in which the material processing is a cutting or welding process.

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